🧮 1. Importing the math module

Before using math functions, you must import the module:

```
import math
```

⚙️ 2. Common math functions

🔢 Basic mathematical operations

```
import math


print(math.sqrt(25))      # Square root → 5.0
print(math.pow(2, 3))     # Power → 2³ = 8.0
print(math.ceil(4.2))     # Ceiling → 5
print(math.floor(4.8))    # Floor → 4
print(math.trunc(4.9))    # Truncate → 4
print(abs(-10))           # Absolute value → 10
```

🔺 Trigonometric functions

All trigonometric functions in math work with **radians** (not degrees).

```
print(math.sin(math.pi / 2))  # sin(90°) = 1.0
print(math.cos(0))            # cos(0°) = 1.0
print(math.tan(math.pi / 4))  # tan(45°) = 1.0


# Convert between radians and degrees
print(math.degrees(math.pi))   # 180.0
print(math.radians(180))      # 3.141592653589793
```

🧭 Logarithmic and exponential functions

```
print(math.log(10))       # Natural log (base e)
print(math.log10(100))    # Log base 10 → 2.0
print(math.exp(2))        # e^2 → 7.389...
```

📏 **Constants**

```
print(math.pi)   # 3.141592653589793
print(math.e)    # 2.718281828459045
print(math.tau)  # 6.283185307179586 (2π)
```

🎲 **Other useful functions**

```
print(math.factorial(5))   # 5! = 120
print(math.gcd(12, 18))    # GCD = 6
print(math.fabs(-7.5))     # 7.5 (float abs)
print(math.isfinite(100))  # True
print(math.isinf(float('inf'))) # True
print(math.isnan(float('nan'))) # True
```

💡 **3. Example Program**

Let's see a practical example that combines multiple math functions:

```
import math

radius = 5
area = math.pi * math.pow(radius, 2)
circumference = 2 * math.pi * radius

print("Radius:", radius)
```

```
print("Area:", round(area, 2))
print("Circumference:", round(circumference, 2))
```

Output:

```
Radius: 5
Area: 78.54
Circumference: 31.42
```

✅ **Summary Table**

Function	Description	Example	Result
math.sqrt(x)	Square root	sqrt(16)	4.0
math.pow(x, y)	Power	pow(2, 3)	8.0
math.log(x)	Natural log	log(10)	2.302...
math.log10(x)	Log base 10	log10(100)	2.0
math.factorial(x)	Factorial	factorial(5)	120
math.gcd(x, y)	Greatest Common Divisor	gcd(8, 12)	4
math.ceil(x)	Round up	ceil(4.2)	5
math.floor(x)	Round down	floor(4.8)	4